# Supplementary Review of: A novel statistical workflow using pollen records and regression kriging to reconstruct the spatially and temporally explicit demographic history of tree species

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## 1 The post-glacial colonization history of *Abies* spp. in Europe

In this study, we reconstructed the full demographic history of three European fir species (*Abies alba*, *Abies borisii-regis*, and *Abies cephalonica*) during the last 20k CalYrBP across Europe in a spatially and temporally explicit manner. Here, we review the findings of this exercise, focusing on the new insights we gained by applying our workflow.

Our reconstruction indicates a continuous expansion of *Abies* spp. from the Early Holocene to approximately 4k CalYrBP, although their RelAb fluctuated over time. Initially, RelAb increased, primarily due to the growth of older populations, before declining across Europe in a northward geographical pattern. In the second half of the Early Holocene, *Abies* populations expanded in the Alps bordering the Po Plain and in the Northern Apennines, consistent with the conclusions of Magri et al. (2015). During this period, Central Europe's climate was likely too continental for *Abies* spp., favoring other species such as *Corylus avellana* L., as suggested by the absence of the drought-sensitive *Fagus sylvatica* and by paleoclimate evidence (Tinner and Lotter, 2001).

Following the Early Holocene, the species' spread and colonization pathways began to diverge. Northern Italian routes split between the western and eastern Alps, as confirmed by genetic data identifying distinct western and eastern ancestral groups in Central Europe (Balao et al., 2024). We also reconstructed the first contact between the Italian and Balkan colonization routes in the Dinaric Alps around 9k CalYrBP. This supports earlier suggestions of a possible contact zone in the area (Liepelt et al., 2009) (but not a refugium). Slightly earlier, around 10k CalYrBP, the Central Apennines were reached. Interestingly, populations from the Southern and Northern Apennines refugia spread in the coastal areas before moving into the Central Apennines, rather than directly through the mountain chain. This region is a known debate point regarding the genetic continuity or potential contact zone between lineages (e.g., Larsen and Mekic, 1991; Scaltsoyiannes et al., 1999; Linares, 2011; Camerano et al., 2012), often contrasting the genetic evidence that suggests a sharp break in the Gran Sasso and Majella massifs (Piotti et al., 2017) or the presence of undetected refugia (Piotti et al., 2017; Major et al., 2021). However, pollen data can underestimate tree population sizes in geographically complex areas, such as the Apennines (Magri et al., 2015), and not all small refugia contribute to the modern gene pool (Magri et al., 2006). Thus, the demographic dynamics of the Central Apennines' populations remain unresolved, mainly due to a scarcity of paleoecological sites. As our workflow is sensitive to data availability (Fig. 4A–C), a future regional study with more evenly distributed mountainous and coastal sites could provide clearer insight into this unresolved question.

In the Middle Holocene, *Abies* spp. RelAb continued to increase in established populations, reaching their maxima at different times across the distribution, in line with local colonization history. This period coincided

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with the Holocene Thermal Maximum, characterized by higher (winter) precipitation and warmer (summer) temperatures than today (up to 2°C) (Samartin et al., 2017; Fischer et al., 2018). These warm and anthropogenically almost undisturbed conditions were likely ideal for *Abies alba*, as demonstrated by Tinner et al., 2013 for the Middle Holocene and other studies of the Eemian period (Zagwijn, 1996; Vitasse et al., 2019; Mazza et al., 2025). During this time, the species expanded into the Carpathians via two main routes: one from Central Europe and one from the Balkan Peninsula, eventually meeting around 4k CalYrBP. This finding is supported by earlier paleoecological work (e.g., Huntley and Birks, 1983) and genetic data identifying a contact zone between northern Italian and Balkan lineages in the Ukrainian and Romanian Carpathians (Liepelt et al., 2002; Liepelt et al., 2009; Gömöry et al., 2012; Bosela et al., 2016).

Finally, in the Late Holocene (∼ 4k CalYrBP), the spread of *Abies* spp. stopped across Europe while simultaneously the RelAb began to decrease in some regions, particularly the Mediterranean peninsulas and the Alps. Other areas, such as the Pannonian Plain and the Carpathians, showed no decline. This Late Holocene decline likely had multiple, region-specific causes, including slow climate change, increased competition, and human disturbance, exacerbated by the interactions among these factors. This could be resolved by integrating proxies for human impact, such as pollen percentages of *Plantago lanceolata* and Cerealia type, into the workflow; however, this would involve applying the method to different pollen types and producing interpolated maps for each indicator independently. Aside from the potential cumulative error, this would require careful consideration due to the region-specific nature of human impact indicators and possible limitations in taxonomic resolution across sites (e.g., distinguishing Cerealia type from other Poaceae in the Mediterranean and Near East is challenging; Lang et al., 2023). In the future, incorporating a covariate that summarizes human impact, e.g., the LUP index explicitly calculated for each site, could be a valuable approach (Deza-Araujo et al., 2022). Overall, our reconstruction of *Abies* presence and abundance aligns well with the modern distribution of *Abies alba*, *Abies cephalonica*, and *Abies borisii-regis*, showing high RelAb in the leading mountain chains of Europe and lower RelAb in Mediterranean refugia and areas historically more accessible to logging.

Looking at the future abundance of European *Abies* spp., based on our workflow, we found only regional decreases in RelAb, especially under the warmer climate scenarios, and increases in the mountainous areas (Fig. 6). While different climate variable sets influenced the final predicted RelAb values, the spatial distribution remained consistent across the scenarios, indicating high RelAb in the current range. These findings are in contrast to those from several studies that predicted a decline of *Abies alba*, the dominant European fir species (Maiorano et al., 2012; Cailleret et al., 2014; Mauri et al., 2016). Conversely, our results agree with those from other studies predicting stable future conditions in the species' current range even under climate warming up to +6°C at the continental scale (Tinner et al., 2013; Ruosch et al., 2016). Unfortunately, the resolution of our workflow does not allow us to disentangle the reasons for a potential increase in population sizes: the suitability of the species to future climates or new colonizable spaces, as rising temperatures open up previously inaccessible (mountainous) areas.

# References


Balao, F., M. Rios, J. M. Sanchez-Robles, M. T. Lorenzo, J. C. Linares, and A. Terrab (2024). "Postglacial Genetic Legacies, Demography, and Climate Change: Setting Conservation Priorities for Silver Fir". In: *BioRxiv*. DOI: 10.1101/2024.10.07.616973.

Bosela, M., I. Popa, D. Gömöry, R. Longauer, B. Tobin, J. Kyncl, T. Kyncl, C. Nechita, R. Petráš, C. G. Sidor, V. Šebeň, and U. Büntgen (2016). "Effects of post-glacial phylogeny and genetic diversity on the growth variability and climate sensitivity of European silver fir". In: *Journal of Ecology* 104.3. Ed. by R. Shefferson, 716–724. ISSN: 1365-2745. DOI: 10.1111/1365-2745.12561.

Cailleret, M., M. Heurich, and H. Bugmann (2014). "Reduction in browsing intensity may not compensate climate change effects on tree species composition in the Bavarian Forest National Park". In: *Forest Ecology and Management* 328, 179–192. ISSN: 0378-1127. DOI: 10.1016/j.foreco.2014.05.030.

Camerano, P., D. Ferrazzini, F. Ducci, and P. Belletti (2012). "Regioni di Provenienza Per l'Abete Bianco". In: *Sherwood* 17, pp. 35–40.

Deza-Araujo, M., C. Morales-Molino, M. Conedera, P. D. Henne, P. Krebs, M. Hinz, C. Heitz, A. Hafner, and W. Tinner (2022). "A new indicator approach to reconstruct agricultural land use in Europe from sedimentary pollen assemblages". In: *Palaeogeography, Palaeoclimatology, Palaeoecology* 599, p. 111051. ISSN: 0031-0182. DOI: 10.1016/j.palaeo.2022.111051.

Fischer, H., K. J. Meissner, A. C. Mix, N. J. Abram, J. Austermann, V. Brovkin, E. Capron, D. Colombaroli, A.-L. Daniau, K. A. Dyez, T. Felis, S. A. Finkelstein, S. L. Jaccard, E. L. McClymont, A. Rovere, J. Sutter, E. W. Wolff, S. Affolter, P. Bakker, J. A. Ballesteros-Cánovas, C. Barbante, T. Caley, A. E. Carlson, O. Churakova, G. Cortese, B. F. Cumming, B. A. S. Davis, A. de Vernal, J. Emile-Geay, S. C. Fritz, P. Gierz, J. Gottschalk, M. D. Holloway, F. Joos, M. Kucera, M.-F. Loutre, D. J. Lunt, K. Marcisz, J. R. Marlon, P. Martinez, V. Masson-Delmotte, C. Nehrbass-Ahles, B. L. Otto-Bliesner, C. C. Raible, B. Risebrobakken,

M. F. Sánchez Goñi, J. S. Arrigo, M. Sarnthein, J. Sjolte, T. F. Stocker, P. A. Velasquez Alvárez, W. Tinner, P. J. Valdes, H. Vogel, H. Wanner, Q. Yan, Z. Yu, M. Ziegler, and L. Zhou (2018). "Palaeoclimate constraints on the impact of 2 °C anthropogenic warming and beyond". In: *Nature Geoscience* 11.7, 474–485. ISSN: 1752-0908. DOI: 10.1038/s41561-018-0146-0.

Gömöry, D., L. Paule, D. Krajmerová, I. Romšáková, and R. Longauer (2012). "Admixture of genetic lineages of different glacial origin: a case study of Abies alba Mill. in the Carpathians". In: *Plant Systematics and Evolution* 298.4, 703–712. ISSN: 1615-6110. DOI: 10.1007/s00606-011-0580-6.

Huntley, B. and H. J. B. Birks (1983). *An atlas of past and present pollen maps for Europe: 0-13000 BP.* Cambridge University Press.

Lang, G., B. Ammann, K.-E. Behre, and W. Tinner (2023). *Quaternary vegetation dynamics of Europe.* Haupt Verlag.

Larsen, J. and F Mekic (1991). "The geographic variation in European silver fir (Abies alba Mill.)" In: *Silvae genetica* 40.5-6, pp. 188–198.

Liepelt, S., R. Bialozyt, and B. Ziegenhagen (2002). "Wind-dispersed pollen mediates postglacial gene flow among refugia". In: *Proceedings of the National Academy of Sciences* 99.22, 14590–14594. ISSN: 1091-6490. DOI: 10.1073/pnas.212285399.

Liepelt, S., R. Cheddadi, J.-L. de Beaulieu, B. Fady, D. Gömöry, E. Hussendörfer, M. Konnert, T. Litt, R. Longauer, R. Terhürne-Berson, and B. Ziegenhagen (2009). "Postglacial range expansion and its genetic imprints in Abies alba (Mill.) — A synthesis from palaeobotanic and genetic data". In: *Review of Palaeobotany and Palynology* 153.1–2, 139–149. ISSN: 0034-6667. DOI: 10.1016/j.revpalbo.2008.07.007.

Linares, J. C. (2011). "Biogeography and evolution of Abies (Pinaceae) in the Mediterranean Basin: the roles of long-term climatic change and glacial refugia: Biogeography and evolution of the circum-Mediterranean firs". In: *Journal of Biogeography* 38.4, 619–630. ISSN: 0305-0270. DOI: 10.1111/j.1365-2699.2010.02458.x.

Magri, D., E. Agrillo, F. Di Rita, G. Furlanetto, R. Pini, C. Ravazzi, and F. Spada (2015). "Holocene dynamics of tree taxa populations in Italy". In: *Review of Palaeobotany and Palynology* 218, 267–284. ISSN: 0034-6667. DOI: 10.1016/j.revpalbo.2014.08.012.

Magri, D., G. G. Vendramin, B. Comps, I. Dupanloup, T. Geburek, D. Gömöry, M. Latałowa, T. Litt, L. Paule, J. M. Roure, I. Tantau, W. O. Van Der Knaap, R. J. Petit, and J. De Beaulieu (2006). "A new scenario for the Quaternary history of European beech populations: palaeobotanical evidence and genetic consequences". In: *New Phytologist* 171.1, 199–221. ISSN: 1469-8137. DOI: 10.1111/j.1469-8137.2006.01740.x.

Maiorano, L., R. Cheddadi, N. E. Zimmermann, L. Pellissier, B. Petitpierre, J. Pottier, H. Laborde, B. I. Hurdu, P. B. Pearman, A. Psomas, J. S. Singarayer, O. Broennimann, P. Vittoz, A. Dubuis, M. E. Edwards, H. A. Binney, and A. Guisan (2012). "Building the niche through time: using 13,000 years of data to predict the effects of climate change on three tree species in Europe". In: *Global Ecology and Biogeography* 22.3, 302–317. ISSN: 1466-8238. DOI: 10.1111/j.1466-8238.2012.00767.x.

Major, E. I., M. Höhn, C. Avanzi, B. Fady, K. Heer, L. Opgenoorth, A. Piotti, F. Popescu, D. Postolache, G. G. Vendramin, and K. Csilléry (2021). "Fine-scale spatial genetic structure across the species range reflects recent colonization of high elevation habitats in silver fir (Abies albaMill.)" In: *Molecular Ecology* 30.20, 5247–5265. ISSN: 1365-294X. DOI: 10.1111/mec.16107.

Mauri, A, D De Rigo, and G Caudullo (2016). *Abies alba in Europe: distribution, habitat, usage and threats.* Publication Office of the European Union Luxembourg, pp. 48–49.

Mazza, G., M. C. Manetti, G. Kraushaar, G. Pezzi, P. Krebs, S. Coşgun, W. Tinner, and M. Conedera (2025). "Climate sensitivity of Abies alba Mill. in marginal Mediterranean low-elevation stands reveals new insights into the ecological potential of the species". In: *Forest Ecology and Management* 585, p. 122670. ISSN: 0378-1127. DOI: 10.1016/j.foreco.2025.122670.

Piotti, A., C. Leonarduzzi, D. Postolache, F. Bagnoli, I. Spanu, L. Brousseau, C. Urbinati, S. Leonardi, and G. G. Vendramin (2017). "Unexpected scenarios from Mediterranean refugial areas: disentangling complex demographic dynamics along the Apennine distribution of silver fir". In: *Journal of Biogeography* 44.7, 1547–1558. ISSN: 1365-2699. DOI: 10.1111/jbi.13011.

Ruosch, M., R. Spahni, F. Joos, P. D. Henne, W. O. van der Knaap, and W. Tinner (2016). "Past and future evolution of Abies alba forests in Europe – comparison of a dynamic vegetation model with palaeo data and observations". In: *Global Change Biology* 22.2, 727–740. ISSN: 1365-2486. DOI: 10.1111/gcb.13075.

Samartin, S., O. Heiri, F. Joos, H. Renssen, J. Franke, S. Brönnimann, and W. Tinner (2017). "Warm Mediterranean mid-Holocene summers inferred from fossil midge assemblages". In: *Nature Geoscience* 10.3, 207–212. ISSN: 1752-0908. DOI: 10.1038/ngeo2891.

Scaltsoyiannes, A., M. Tsaktsira, and A. D. Drouzas (1999). "Allozyme differentiation in the Mediterranean firs (Abies, Pinaceae). A first comparative study with phylogenetic implications". In: *Plant Systematics and Evolution* 216.3–4, 289–307. ISSN: 1615-6110. DOI: 10.1007/bf01084404.

Tinner, W., D. Colombaroli, O. Heiri, P. D. Henne, M. Steinacher, J. Untenecker, E. Vescovi, J. R. M. Allen, G. Carraro, M. Conedera, F. Joos, A. F. Lotter, J. Luterbacher, S. Samartin, and V. Valsecchi (2013).

"The past ecology of Abies alba provides new perspectives on future responses of silver fir forests to global warming". In: *Ecological Monographs* 83.4, 419–439. ISSN: 1557-7015. DOI: 10.1890/12-2231.1.

Tinner, W. and A. F. Lotter (2001). "Central European vegetation response to abrupt climate change at 8.2 ka". In: *Geology* 29.6, pp. 551–554.

Vitasse, Y., A. Bottero, M. Rebetez, M. Conedera, S. Augustin, P. Brang, and W. Tinner (2019). "What is the potential of silver fir to thrive under warmer and drier climate?" In: *European Journal of Forest Research* 138.4, 547–560. ISSN: 1612-4677. DOI: 10.1007/s10342-019-01192-4.

Zagwijn, W (1996). "An analysis of Eemian climate in Western and Central Europe". In: *Quaternary Science Reviews* 15.5–6, 451–469. ISSN: 0277-3791. DOI: 10.1016/0277-3791(96)00011-x.